TDP-Reasoner: Timing-Dependent Plasticity Along the Recursion

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Abstract

Small recursive models trained from scratch are the strongest non-LLM approach to puzzle benchmarks such as Sudoku-Extreme and ARC-AGI. These models, HRM and TRM among them, re-apply a Transformer block many times. Seen as a modern Hopfield network, each step's attention is a Hebbian memory of the current state, so recursion reinforces the model's current answer, and accuracy falls once it runs past its training horizon. We propose TDP-Reasoner (timing-dependent plasticity reasoner), which keeps a fast-weight memory along the recursion itself: keys and values from different steps are paired under a learned two-sided timing window, borrowed from spike-timing-dependent plasticity, so the memory tracks how the state moves rather than where it is. Demonstrations of a task reach the query only through a per-task linear rule fitted in context, so unseen tasks need no retraining. Trained for 256 recursion blocks and run for 10,000, it raises held-out Sudoku-Extreme accuracy from 59.8% to 82.0% and never loses a solved puzzle. On unseen ARC-AGI-1 tasks with small grids it matches the public TRM checkpoint (43.2% vs. 45.5%) after about 3% of its training; at full scale it reaches 18.8% pass@2 on ARC-AGI-1 (TRM: 41.8% under the same evaluation) and 0% on ARC-AGI-2.